Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

Learning goal: I can use a simulation to find experimental probability. (circle your level after scoring)

4	I can design a simulation for a situation and predict results for a larger number of trials.
3	I can use results from a simulation to find an experimental probability and compare it to theoretical.
2	I can record the results of trials in a table.
1	I need help to start this benchmark.

1. A fair 12-sided die is rolled 300 times. It lands on a 9 exactly 24 times.

Part A. What is the experimental probability of rolling a 9? Write it as a fraction in simplest form.
Part B. What is the theoretical probability of rolling a 9? Write it as a fraction in simplest form.
Part C. Is the experimental probability greater than or less than the theoretical probability?

2. A fair coin is tossed 400 times and lands on heads 204 times.

What is the experimental probability of landing on heads? Write your answer as a decimal.

3. Two students each simulate rolling a fair die. One runs 5,900 trials and the other runs 370 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. The 5,900-trial student, because more trials usually bring the result closer to the theory.

B. Both students equally, because a fair experiment gives the same result for any number of trials.

C. The 370-trial student, because fewer trials leave fewer chances for a result to go wrong.

D. Neither student, because an experimental result never comes close to the theoretical one.

Turn over →

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____

4. A spinner has 10 equal sections, and 8 of them are red. In a simulation the spinner is spun 400 times and lands on red 311 times.

Which statement correctly compares the experimental and theoretical probabilities of landing on red?

A. The experimental probability, $\frac{311}{400}$, is a little less than the theoretical probability, $\frac{4}{5}$.

B. The theoretical probability is $\frac{311}{400}$, because that is what actually happened.

C. The experimental probability, $\frac{311}{400}$, is a little greater than the theoretical probability, $\frac{4}{5}$.

D. The two probabilities must be equal because the spinner is fair.

5. A bag holds 10 marbles. Each marble is green, purple or blue, but the number of each color is not known. A student draws a marble at random, records its color and puts it back. The table shows the results of 200 draws.

Green	Purple	Blue
121	43	36

Part A. What is the experimental probability of drawing a purple marble? Write it as a decimal.
Part B. Use your answer to Part A. If the student makes 2,000 more draws, about how many times should a purple marble be drawn?
Part C. About how many of the 10 marbles in the bag are purple?

Score: ____ / 5 **My level: 1 2 3 4** **→ copy both to your progress strip**

Answer key — C-DP.2.4 · MA.7.DP.2.4 · Experimental vs theoretical probability

Unit 2: Probability. Regular = on-benchmark item; Rigor = the item that needs the benchmark's reasoning. Mastery on the weekly check = 4 of 5.

Weekly check (5 items)

1. Regular **Experimental: $\frac{2}{25}$; theoretical: $\frac{1}{12}$. The experimental probability is slightly less than the theoretical probability. Score: 1 point for all three parts; the comparison alone earns 0.**

Part A. Experimental probability = times it happened ÷ trials = $\frac{24}{300} = \frac{2}{25} = 0.08$.

Part B. Theoretical probability = $1 \text{ favorable face} \div 12 \text{ faces} = \frac{1}{12} \approx 0.0833$.

Part C. The experimental probability is slightly less than the theoretical probability. They are close but not equal; with more trials the experimental probability usually gets closer to $\frac{1}{12}$.

2. Regular **0.51**

Experimental probability = successes ÷ trials.

$204 \div 400 = 0.51$.

3. Regular **A. The 5,900-trial student, because more trials usually bring the result closer to the theory.**

Experimental probabilities vary from trial to trial, but as the number of trials increases they typically get closer to the theoretical probability.

The 5,900-trial simulation is expected to be closer.

4. Regular **A. The experimental probability, $\frac{311}{400}$, is a little less than the theoretical probability, $\frac{4}{5}$.**

Theoretical: $\frac{8}{10} = \frac{4}{5} = 0.8$.

Experimental: $\frac{311}{400} = 0.7775$.

The experimental probability is a little less than the theoretical probability because of random variation.

5. Rigor **Part A: 0.215. Part B: about 430 times. Part C: about 2 purple marbles. Score: 1 point for all three parts; Part A alone earns 0.**

Part A. Experimental probability = purple draws ÷ all $\text{draws} = \frac{43}{200} = 0.215$.

Part B. The bag has not changed, so about the same part of the new draws should be purple: $0.215 \times 2{,}000 = 430$ times. (The actual count will vary a little.)

Part C. The experimental probability estimates the theoretical probability, which is the number of purple marbles out of 10: $0.215 \times 10 = 2.15$, so about 2 of the marbles are purple ($\frac{2}{10} = 0.2$, close to 0.215).

Re-check strip (3 items, C-DP.2.4-R)

1. Regular **B. The 8,000-trial student, because more trials usually bring the result closer to the theory.**

Experimental probabilities vary from trial to trial, but as the number of trials increases they typically get closer to the theoretical probability.

The 8,000-trial simulation is expected to be closer.

2. Regular **0.47**

Experimental probability = successes ÷ trials.

$376 \div 800 = 0.47$.

3. Regular **about 100 times**

The theoretical probability of drawing purple is 0.25, so about 25% of the draws should be purple.

$0.25 \times 400 = 100$.

About 100 of the 400 draws would be expected to be purple. The actual number will usually differ a little, because of random variation.

Comparison strip (2 items, C-DP.2.4-X) — used on a later week's re-check when this benchmark was *not* retaught

1. Regular **C. The 4,700-trial student, because more trials usually bring the result closer to the theory.**

Experimental probabilities vary from trial to trial, but as the number of trials increases they typically get closer to the theoretical probability.

The 4,700-trial simulation is expected to be closer.

2. Regular **Experimental: $\frac{23}{200}$; theoretical: $\frac{1}{10}$. The experimental probability is slightly greater than the theoretical probability. Score: 1 point for all three parts; the comparison alone earns 0.**

Part A. Experimental probability = times it happened ÷ trials = $\frac{23}{200} = 0.115$.

Part B. Theoretical probability = $1 \text{ favorable face} \div 10 \text{ faces} = \frac{1}{10} = 0.1$.

Part C. The experimental probability is slightly greater than the theoretical probability. They are close but not equal; with more trials the experimental probability usually gets closer to $\frac{1}{10}$.

Reteach plan — C-DP.2.4

MA.7.DP.2.4 · Experimental vs theoretical probability

Item that gave the most trouble (from both tally sheets):

What the mistake looks like:

Worked example for Tuesday (8–10 min):

Three practice items (one change at a time):

Decision line: Reteach on _____ Huddle: in person / paper Date: _____ Re-check given on _____

7th Grade Math · Re-check (3 items) C-DP.2.4-R

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate spinning a fair spinner. One runs 210 trials and the other runs 8,000 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. The 210-trial student, because fewer trials leave fewer chances for a result to go wrong.

B. The 8,000-trial student, because more trials usually bring the result closer to the theory.

C. Both students equally, because a fair experiment gives the same result for any number of trials.

D. Neither student, because an experimental result never comes close to the theoretical one.

2. A fair coin is tossed 800 times and lands on heads 376 times.

What is the experimental probability of landing on heads? Write your answer as a decimal.

3. A bag contains only purple marbles and blue marbles. When one marble is drawn at random, the probability that it is purple is 0.25. A student draws a marble, records its color and returns it to the bag. The student repeats this 400 times.

About how many times would you expect the student to draw a purple marble?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Re-check (3 items) C-DP.2.4-R

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate spinning a fair spinner. One runs 210 trials and the other runs 8,000 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

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C. Both students equally, because a fair experiment gives the same result for any number of trials.

D. Neither student, because an experimental result never comes close to the theoretical one.

2. A fair coin is tossed 800 times and lands on heads 376 times.

What is the experimental probability of landing on heads? Write your answer as a decimal.

3. A bag contains only purple marbles and blue marbles. When one marble is drawn at random, the probability that it is purple is 0.25. A student draws a marble, records its color and returns it to the bag. The student repeats this 400 times.

About how many times would you expect the student to draw a purple marble?

Score: ____ / 3 → **copy to your progress strip**

7th Grade Math · Comparison items (2)

C-DP.2.4-X

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate rolling a fair die. One runs 170 trials and the other runs 4,700 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. Neither student, because an experimental result never comes close to the theoretical one.

B. The 170-trial student, because fewer trials leave fewer chances for a result to go wrong.

C. The 4,700-trial student, because more trials usually bring the result closer to the theory.

D. Both students equally, because a fair experiment gives the same result for any number of trials.

2. A fair 10-sided die is rolled 200 times. It lands on a 3 exactly 23 times.

Part A. What is the experimental probability of rolling a 3? Write it as a fraction in simplest form.
Part B. What is the theoretical probability of rolling a 3? Write it as a fraction in simplest form.
Part C. Is the experimental probability greater than or less than the theoretical probability?

Score: ____ / 2 → copy to your progress strip

7th Grade Math · Comparison items (2)

C-DP.2.4-X

Benchmark: **MA.7.DP.2.4** · Experimental vs theoretical probability Student code: _____ Calculator: yes

1. Two students each simulate rolling a fair die. One runs 170 trials and the other runs 4,700 trials.

Whose experimental probability is expected to be closer to the theoretical probability?

A. Neither student, because an experimental result never comes close to the theoretical one.

B. The 170-trial student, because fewer trials leave fewer chances for a result to go wrong.

C. The 4,700-trial student, because more trials usually bring the result closer to the theory.

D. Both students equally, because a fair experiment gives the same result for any number of trials.

2. A fair 10-sided die is rolled 200 times. It lands on a 3 exactly 23 times.

Part A. What is the experimental probability of rolling a 3? Write it as a fraction in simplest form.
Part B. What is the theoretical probability of rolling a 3? Write it as a fraction in simplest form.
Part C. Is the experimental probability greater than or less than the theoretical probability?

Score: ____ / 2 → copy to your progress strip